Thesis -> Paper

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Abstract

This thesis is being transformed into a paper

1 Introduction

1.1 Motivation and Research Context

The mean-variance framework introduced by Markowitz [1952] has shaped capital allocation for over seven decades, yet its core assumptions of stable correlations and reliable parameter estimates break down in exactly the conditions where diversification is meant to help. During market declines, correlations among S&P 500 constituents rise well above their upside levels, an asymmetry that erodes the diversification the framework is built to deliver precisely when protection matters most [Ang and Chen 2002].

2 Methodology

2.1 Time Frame

- 28 years, January 1998 untill December 2025

2.2 Rebalancing

- Rebalancing Schedule 3x, monthly, quarterly, yearly - if non trading day, next day in the month/year

2.3 Weight Constraints

- 1% min and 10% maximum allocation per stock per rebalancing date. - Intra rebalancing drift can exceed 10%

2.4 Lookback Windows

- 60 month calculation window for HRP and MVO - Fixed 60 month lookback window for all Machine Learning models | 24 month validation | + 1, 3, 12 month embargo

2.5 Coverage and Missing Data

- price data of min. 50% given, otherwise excluded - 5 days forward filled, otherwise missing values

2.6 Transaction Cost

- 0bps, 10bps, 20bps calculated. 10bps standard transaction cost - Turnover is getting calculated and reported

2.7 Point-in-Time

- Everything is point in time only

2.8 Features

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3 Regime Implementation

3.1 Changepoint Detection

3.1.1 Step 1

- daily VIX log changes - k-means clustering and mood rank test. is the spread of teh data after k different from before k? -> keeps the largest test statistic - if that maximum exceeds a threshold, it raises an alarm on day t (detection date). the k with the largest statistic becomes teh estimated break date t, search restarts from there - ARL0 = 10000 - Mood test uses ranks, it assumes no distribution and it reacts to changes in volatility

3.1.2 Step 2

- estimate teh volatility once. - each detection day t, an EWMA (λ = 0.95) of squared S&P 500 returns from t det to t gives the new regime's annualised volatility o. Always SPY returns, because 20% annual volatility is from there - Freeze it. o stays constant until the new detection. That is why your chart has steps and late exits: calm returns only after a new break is detected and the new segment's vol is below 20- Map it to a score: p$_c risis$ =Φ((σ − 0.20) / 0.05), $where$Φ$isthestandardnormalCDF.Soo =$ $15\%gives0.16, 20\%gives0.50, 25\%gives0.84andetc.-shiftbyoneday.thevalueondaydusesdataonlythrou$ $1, whichmathcesthepaper'sonedayimplementationdelay$

3.2 Hidden Markov Model

3.2.1 Step 1

- daily log return of S&P 500 - states follow a markov chain. false L holds the probabilities of staying in each state - never the state, only returns. the model infers the state from how well each day's return fits each state's distribution - crisis is defined as teh state with teh hgher o.

3.2.2 Step 2

3.3 Wasserstein K-Means Clustering

4 Regimes

4.1 Changepoint Detection

Methodology and calculation 1:1 to the paper

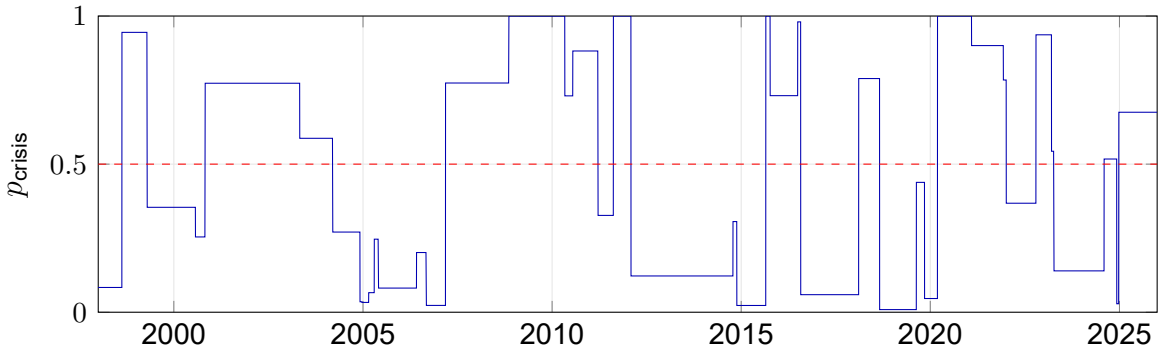


Figure 1: Changepoint detection (VIX): crisis probability p_{crisis}.

4.2 HMM

4.3 Wasserstein

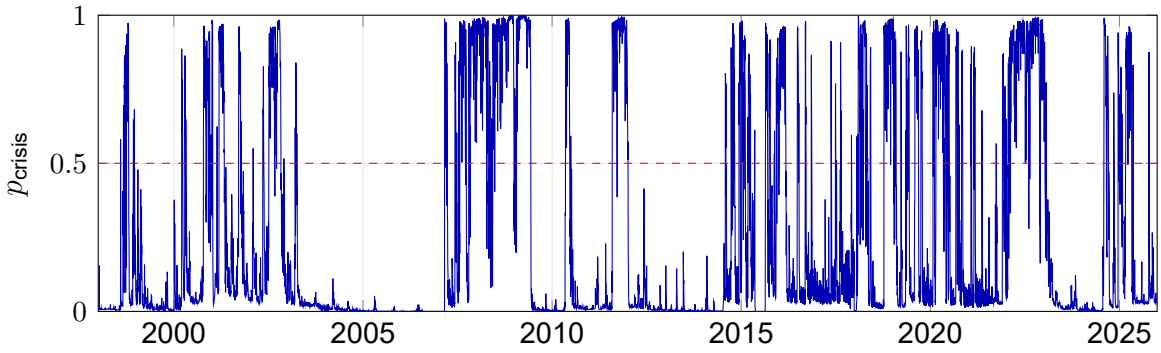


Figure 2: Hidden Markov Model (S&P 500): crisis probability p_{crisis}.

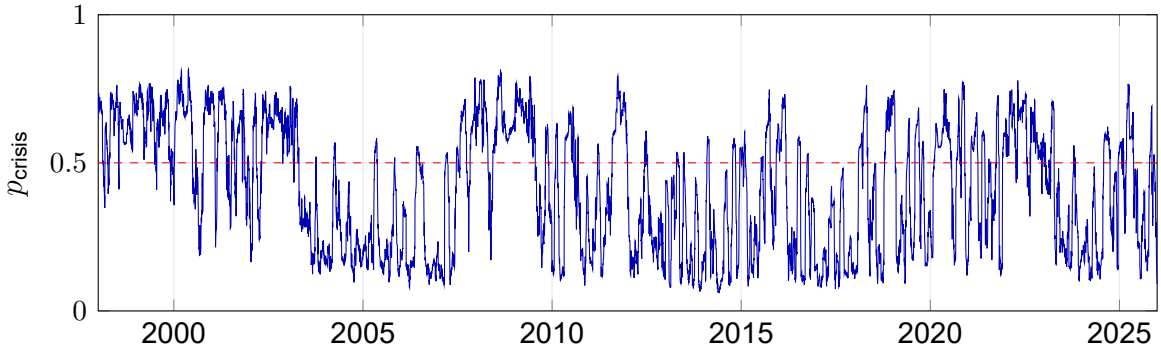


Figure 3: Wasserstein clustering (S&P 500 and VIX): crisis probability p_{crisis}.

5 References

Nystup Peter, Madsen Hendrik, Lindström Erik, 2017. Dynamic portfolio optimization across hidden market regimes.
Used for:
- Transaction costs 10bps
- Hidden Markov Model